

# Deploying IBM Guardium Cryptography Manager

A staged deployment model for Guardium  
Cryptography Manager, Quantum Safe  
Explorer, and Quantum Safe Remediator



## Highlights

Initial quantum-safe  
readiness assessment

Expanded cryptographic  
coverage and context

Remediation and  
cryptographic posture  
management

Immediate quantum risk  
mitigation

Organizations are struggling to manage their cryptography and protect their sensitive data. Quantum computing advancements are ushering in new threats and regulations. With the advancement of IBM® Guardium® quantum safe software, IBM Guardium Cryptography Manager, IBM Quantum Safe Explorer, and IBM Quantum Safe Remediator help your organization achieve crypto-agility, encrypt sensitive data and prepare for PQC readiness.

This staged deployment model provides a practical, value-driven roadmap to help organizations on the journey of quantum-safe migration. With this model, organizations can start quickly, see immediately value, and expand over time as you transition to quantum-safe cryptography. Each stage stands on its own while naturally enabling the next, guiding organizations through a transformation towards crypto-agility and quantum resilience.

| Cryptographic results ⓘ                        |                  |                                                                                                                                         | 331<br>Lines | 5<br>Files | 39<br>Methods |
|------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------|------------|---------------|
| Language<br><div>Java</div>                    |                  |                                                                                                                                         |              |            |               |
| Cryptographic inventory                        |                  |                                                                                                                                         |              |            |               |
| Function Name                                  | Library          | Location                                                                                                                                |              |            |               |
| Filter functions names                         | Filter libraries | Filter locations                                                                                                                        |              |            |               |
| java.security.KeyPairGenerator.getInstance     | JCA              | C:\Varun_Data\Quantum_Safe\vsc-extension\Analytics\Scannable\EncryptedChat\src\main\java\cryptchat\server\dh\KeyExchange.java Line : 51 |              |            |               |
| java.security.KeyPairGenerator.initialize      | JCA              | C:\Varun_Data\Quantum_Safe\vsc-extension\Analytics\Scannable\EncryptedChat\src\main\java\cryptchat\server\dh\KeyExchange.java Line : 52 |              |            |               |
| java.security.KeyPairGenerator.generateKeyPair | JCA              | C:\Varun_Data\Quantum_Safe\vsc-extension\Analytics\Scannable\EncryptedChat\src\main\java\cryptchat\server\dh\KeyExchange.java Line : 54 |              |            |               |
| javax.crypto.KeyAgreement.getInstance          | JCA              | C:\Varun_Data\Quantum_Safe\vsc-extension\Analytics\Scannable\EncryptedChat\src\main\java\cryptchat\server\dh\KeyExchange.java Line : 68 |              |            |               |
|                                                |                  | C:\Varun_Data\Quantum_Safe\vsc-                                                                                                         |              |            |               |

Figure 1. IBM Quantum Safe Explorer scan results

## Stage 1: Initial Quantum-Safe Readiness Assessment

### Scope

Networks and source code

### Goal

Establish an immediate view of quantum-safe readiness and cryptographic posture to reduce security and compliance risk

### Key Capabilities

- Network cryptographic discovery with Guardium Cryptography Manager’s built-in scanner and the out-of-the-box Qualys/Nessus integrations
- Source code cryptographic analysis with Quantum Safe Explorer
- Automatic policy evaluation and quantum-safe readiness assessment in Guardium Cryptography Manager
- Unified dashboards showing quantum-safe readiness and overall cryptographic posture

### Value Delivered

- Time-to-value in days
- Actionable, prioritized initial readiness posture

| Name                                 | Type     | Linked IT assets | Combined violation severity |
|--------------------------------------|----------|------------------|-----------------------------|
| kushaq.dev.fyre.ibm.com:44413-tls-37 | Protocol | 1 Service        | Critical                    |
| kushaq.dev.fyre.ibm.com:44403-tls-50 | Protocol | 1 Service        | Critical                    |
| kushaq.dev.fyre.ibm.com:44406-tls-44 | Protocol | 1 Service        | Critical                    |
| kushaq.dev.fyre.ibm.com:44405-tls-39 | Protocol | 1 Service        | Critical                    |
| kushaq.dev.fyre.ibm.com:44415-tls-36 | Protocol | 1 Service        | Critical                    |
| kushaq.dev.fyre.ibm.com:44412-tls-45 | Protocol | 1 Service        | Critical                    |

Figure 2. IBM Guardium Cryptography Manager inventory view

## Stage 2: Expanded Cryptographic Coverage and Context

### Scope

Enterprise-wide cryptographic landscape and business context

### Goal

Complete the picture to further reduce security and compliance risk and prioritize remediation tasks according to business criticality

### Key Capabilities

- Additional network scanner integrations
- Ingestion of Cryptography Bill of Materials (CBOMs) from any source that outputs CBOMs
- System integrations such as Vault PKI, HSMs, key management systems, and CMDBs

### Value Delivered

- Enterprise-wide quantum-safe readiness posture
- Business-relevant prioritization of cryptographic risk
- Strong foundation for audits, reporting, and roadmap planning

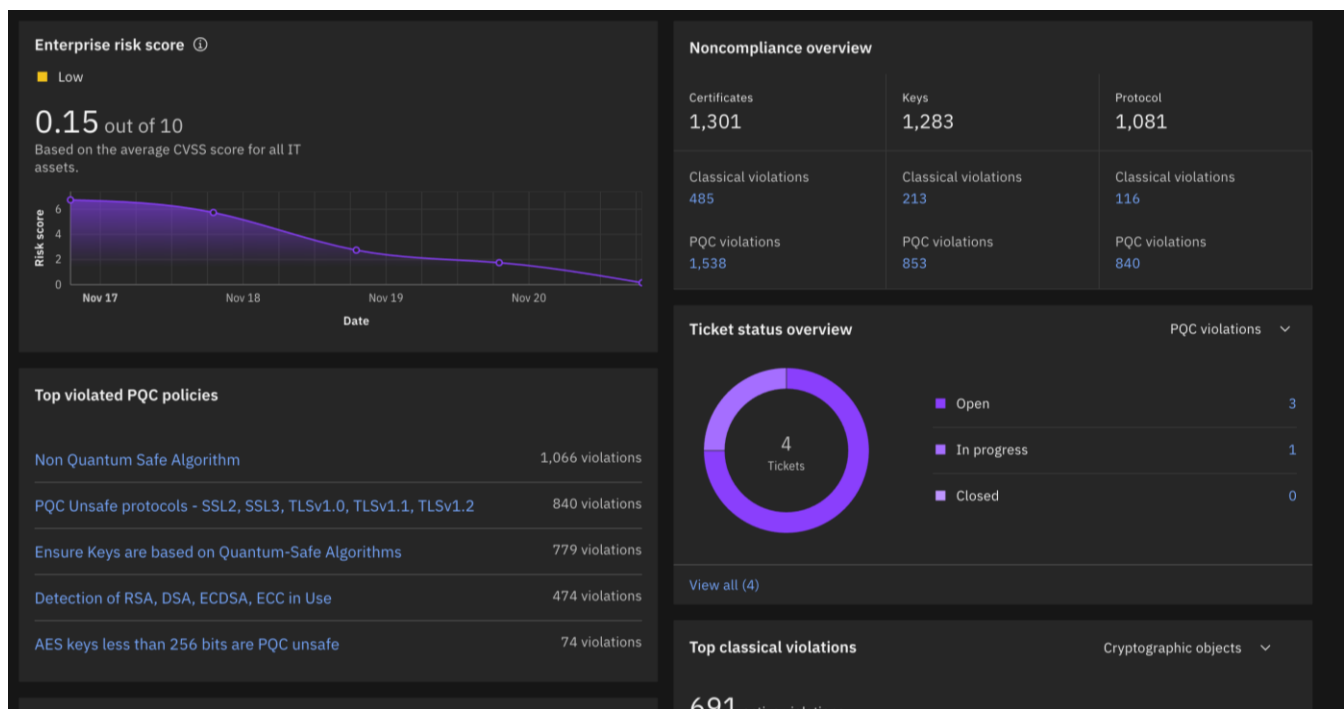


Figure 3. IBM Guardium Cryptography Manager dashboard

## Stage 3: Remediation and Cryptographic Posture Management

### Scope

Full cryptographic landscape

### Goal

Fix and govern cryptography at scale to reduce security and compliance risk, and to improve crypto-agility

### Key Capabilities

- Ticket-based remediation via ServiceNow and Jira integrations
- Prescriptive, line-of-code fixes from Quantum Safe Explorer
- Certificate lifecycle management via Vault PKI integration
- Centralized Transparent Database Encryption (TDE) key management for database systems

### Value Delivered

- Reduced remediation time and operational risk
- Continuous compliance with cryptographic standards and mandates
- Improved crypto-agility

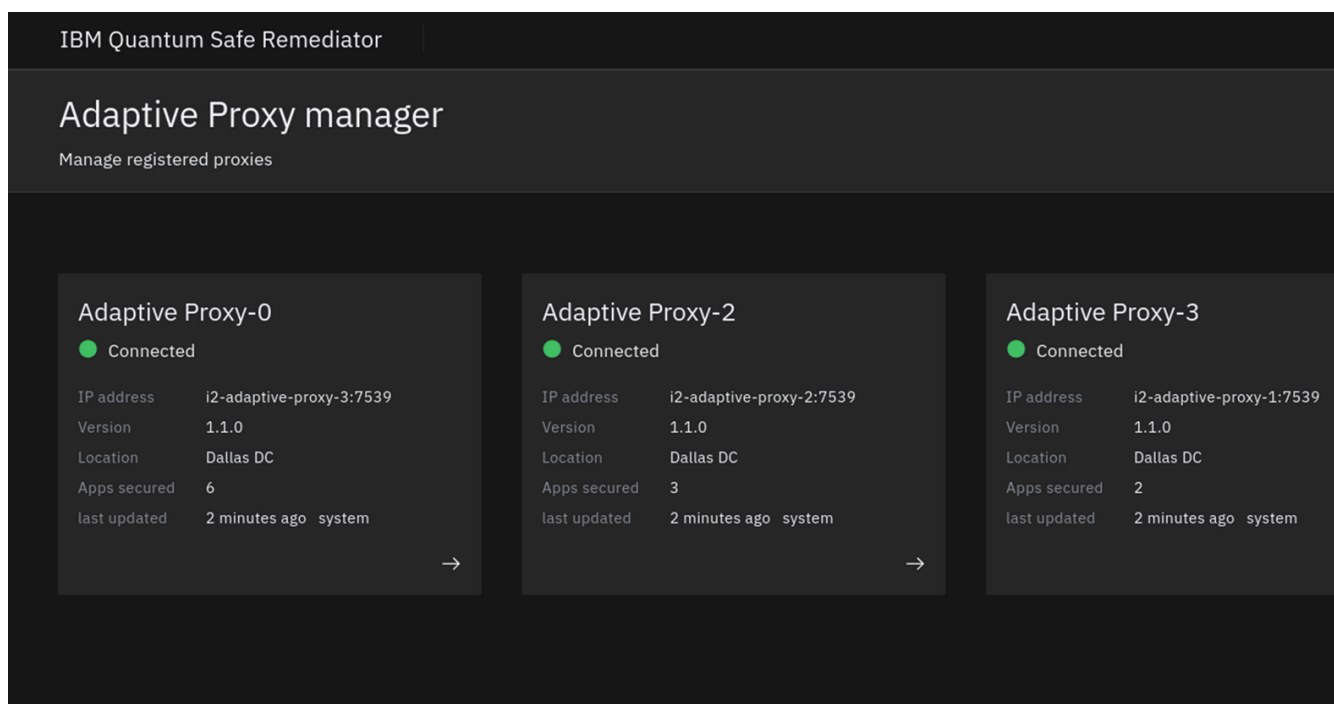


Figure 4. IBM Quantum Safe Remediator Adaptive Proxy manager

## Stage 4: Immediate Quantum Risk Mitigation

### Scope

TLS channels

### Goal

Defend against “harvest now, decrypt later” attacks and enhance crypto-agility as quantum-safe migration continues

### Key Capabilities

- Quantum Safe Remediator reverse proxy
- Quantum Safe Remediator forward proxy

### Value Delivered

- Immediate protection against “harvest now, decrypt later” threats
- Mitigation of applications where quantum-safe migration is not feasible or desirable
- Enhanced crypto-agility

**For more information**

To learn more about Guardium Cryptography Manager, Quantum Safe Explorer, and Quantum Safe Remediator, contact your IBM representative or IBM Business Partner, or visit <https://www.ibm.com/products/guardium-cryptography-manager>.

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